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MUT ICT PORTAL - DATABASE SETUP

INSTRUCTIONS

MUT ICT PORTAL

MySQL Database Setup Instructions

PART 1: WHAT YOU NEED BEFORE STARTING

- MySQL installed on your computer

- MySQL root password
- Terminal/Command Prompt access

PART 2: COMPLETE DATABASE SETUP SCRIPT

Step 1: Open Terminal (Mac/Linux) or Command Prompt (Windows)

Step 2: Type the following command and press Enter:

```
text  
  
mysql -u root -p
```

Step 3: Enter your MySQL password when prompted

Step 4: Copy the entire SQL script below and paste it into MySQL, then press Enter:

```
sql  
  
DROP DATABASE IF EXISTS mut_ict_portal;  
CREATE DATABASE mut_ict_portal;  
USE mut_ict_portal;  
  
CREATE TABLE users (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    student_id VARCHAR(20) UNIQUE NOT NULL,  
    full_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    phone VARCHAR(15),  
    program VARCHAR(100),  
    year_level VARCHAR(5),  
    password VARCHAR(255) NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP  
);
```

```

CREATE TABLE events (
    id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(200) NOT NULL,
    description TEXT,
    event_date DATE NOT NULL,
    event_month VARCHAR(20),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE bursaries (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    description TEXT,
    application_url VARCHAR(255),
    deadline DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

INSERT INTO events (title, description, event_date, event_month) VALUES
('Registration Day', 'All new students register for programs', '2026-01-15', 'January'),
('Freshers Party', 'Welcome event for first-year students', '2026-02-10', 'February'),
('Tech Innovation Day', 'Showcase of student projects', '2026-03-20', 'March'),
('Career Fair', 'Meet potential employers', '2026-06-05', 'June'),
('Hackathon', 'Collaborative coding event', '2026-08-15', 'August'),
('Graduation Day', 'Celebrate student achievements', '2026-11-30', 'November');

INSERT INTO bursaries (name, description, application_url, deadline)
VALUES
('NSFAS', 'Government funding for tuition and living expenses',
'https://www.nsfas.org.za', '2026-01-31'),
('Merit Bursary', 'For students with excellent academic performance',
'https://www.motsepefoundation.org', '2026-02-15'),
('INSETA Sponsorship', 'Corporate sponsorship for ICT students',
'https://inseta.org.za/issf/', '2026-03-30');

INSERT INTO users (student_id, full_name, email, phone, program,
year_level, password) VALUES
('20240001', 'Thabo Student', 'student@mut.ac.za', '0712345678', 'Diploma
in IT', '2',
'$2a$10$92IXUNpkj00r0Q5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi');

```

```
SELECT 'DATABASE SETUP COMPLETE!' AS Message;
SELECT 'Test Login - Student ID: 20240001, Password: password123' AS
Credentials;
```

Step 5: Type EXIT; and press Enter to close MySQL

PART 3: WHAT THE SCRIPT CREATES

Item	Details
Database Name	mut_ict_portal
Tables Created	users, events, bursaries
Test User	1 account (Student ID: 20240001, Password: password123)
Events	6 sample events
Bursaries	3 sample funding options

PART 4: CONNECTION FILE FOR NODE.JS

Step 1: Create a new file in your project folder called db.js

Step 2: Copy and paste the code below into db.js:

```
javascript
```

```
const mysql = require('mysql2');

const pool = mysql.createPool({
  host: 'localhost',
  user: 'root',
  password: 'YOUR_MYSQL_PASSWORD',
  database: 'mut_ict_portal',
  waitForConnections: true,
  connectionLimit: 10,
  queueLimit: 0
});

const promisePool = pool.promise();
module.exports = promisePool;
```

Step 3: Change YOUR_MYSQL_PASSWORD to your actual MySQL root password

Step 4: Save the file

PART 5: INSTALL MYSQL DRIVER

Open terminal in your project folder and run:

```
bash

npm install mysql2
```

PART 6: VERIFY DATABASE IS WORKING

Step 1: Open MySQL again:

```
bash

mysql -u root -p
```

Step 2: Run these commands to verify:

```
sql
USE mut_ict_portal;
SHOW TABLES;
SELECT COUNT(*) FROM users;
SELECT COUNT(*) FROM events;
SELECT COUNT(*) FROM bursaries;
EXIT;
```

Expected Results:

- users table: 1 record
- events table: 6 records
- bursaries table: 3 records

PART 7: BACKUP YOUR DATABASE

To create a backup file:

```
bash
mysqldump -u root -p mut_ict_portal > mut_ict_portal_backup.sql
```

To restore from backup:

```
bash
mysql -u root -p mut_ict_portal < mut_ict_portal_backup.sql
```

PART 8: USEFUL COMMANDS FOR REFERENCE

Command

Purpose

<code>mysql -u root -p</code>	Login to MySQL
<code>SHOW DATABASES;</code>	List all databases
<code>USE mut_ict_portal;</code>	Select the database
<code>SHOW TABLES;</code>	Show all tables
<code>SELECT * FROM users;</code>	View all users
<code>SELECT * FROM events;</code>	View all events
<code>SELECT * FROM bursaries;</code>	View all bursaries
<code>DROP DATABASE mut_ict_portal;</code>	Delete the database
<code>EXIT;</code>	Exit MySQL

PART 9: TROUBLESHOOTING

Problem: "Access denied for user"

Solution: You entered wrong password. Try again or reset MySQL password

Problem: "Can't connect to MySQL server"

Solution: MySQL service is not running. Start it:

- Windows: Search "Services" → Find MySQL → Start
- Mac: `brew services start mysql`
- Linux: `sudo systemctl start mysql`

Problem: "Unknown database 'mut_ict_portal'"

Solution: Database was not created. Run the setup script again

Problem: "Table already exists"

Solution: Run `DROP DATABASE mut_ict_portal;` first, then run setup script

PART 10: DATABASE STRUCTURE DETAILS

Users Table Columns:

- id - Auto-numbered primary key
- student_id - Student's ID number (unique)
- full_name - Student's complete name
- email - Email address (unique)
- phone - Contact phone number
- program - Enrolled program name
- year_level - Year of study (1, 2, or 3)
- password - Encrypted password
- created_at - Registration date/time
- updated_at - Last update date/time

Events Table Columns:

- id - Auto-numbered primary key
- title - Event name
- description - Event details
- event_date - Date of the event
- event_month - Month name
- created_at - Record creation date

Bursaries Table Columns:

- id - Auto-numbered primary key
- name - Bursary name
- description - Bursary details
- application_url - Link to apply
- deadline - Application deadline
- created_at - Record creation date

SETUP COMPLETE

Your MySQL database is now ready to use with the MUT ICT Portal application.